Chirality detected by the order of one word

Human Compression project

We give finite certificates for twelve sporadic cases of the existing chiral-polyhedron existence theorem of Leemans and Liebeck [1]. The same short word detects chirality in every case. The two Mathieu pairs printed in that paper are retained verbatim; the other pairs are explicit new witnesses to already known existence assertions. Six of their ambient representations are acquired from our earlier, separately reviewed certificates. None of these cases is claimed as a new existence theorem or as a solution to an open conjecture.

1 The common criterion

If w is a word in two letters, write $w(x,t)$ for its value at x,t.

Proposition 1. *Suppose $G=\langle x,t\rangle$ is a finite group, t has order two, and some word w satisfies*

$$|w(x,t)|\neq|w(x^{-1},t)|.$$

No automorphism of G can invert x and fix t.

Proof. An automorphism with this action would send $w(x,t)$ to $w(x^{-1},t)$. An isomorphism preserves element orders, contradicting the displayed inequality. This excludes all automorphisms with the stated action, including the involutory ones in the geometric criterion. □

The proposition is an elementary use of word invariants, not a new automorphism theorem. It is useful because a single checked inequality replaces a search through the automorphism group.

For completeness, suppose also that $Z(G)=1$, and put $y=x^{-1}t$. An element of $\langle x\rangle\cap\langle y\rangle$ commutes with both generators x,y, and hence belongs to $Z(G)$. Thus

$$\langle x\rangle\cap\langle y\rangle=1,\qquad (xy)^2=1.$$

In all the examples below $|x|,|y|>2$. These are precisely the rank-three rotation intersection and product conditions. We retain the geometric reconstruction theorem of Schulte and Weiss [2]: these data give a directly regular or chiral polyhedron, and the automorphism in Proposition 1 is the regularity alternative. Its absence gives a chiral polyhedron of type $\{|x|,|y|\}$ with full automorphism group G. The sporadic groups below are nonabelian simple groups, so their centres are trivial. Their identification and simplicity are explicit retained classification inputs.

2 Twelve exact instances

In every row use the single word

$$w(x,t)=xtx^2tx^3t.$$

The last two columns differ in every case.

G	$\lvert G\rvert$	$\lvert x\rvert$	$\lvert x^{-1}t\rvert$	$\lvert w(x,t)\rvert$	$\lvert w(x^{-1},t)\rvert$
M_{11}	7920	11	8	8	3
M_{12}	95040	8	8	10	11
M_{22}	443520	8	7	7	5
M_{23}	10200960	11	15	14	6
M_{24}	244823040	23	15	15	12
J_1	175560	10	6	6	19
J_2	604800	15	15	10	15
J_3	50232960	10	9	4	17
HS	44352000	8	7	7	11
He	4030387200	14	15	17	21
Suz	448345497600	10	9	12	8
McL	898128000	30	8	14	11

These rows assert existence. They do not recover the source's ancillary counts of nonisomorphic polyhedra or require its particular example type when we supply a different witness.

The files `mathieu_pairs.g`, `rank4_pairs.g` and `small_pairs.g` specify every permutation. For M_{23}, M_{24}, the source gives x, y and we set $t = xy$. The checker tests the two permutations literally and computes their orders. For M_{23} it checks their actual membership in the named Mathieu group. For M_{24} the displayed labelling differs from the installed standard one, so the checker constructs a bijective homomorphism onto that named group. Equality of orders alone is not used to identify it.

For $M_{12}, J_2, HS, He, Suz, J_3$, let (r_1, r_2, r_3, r_4) be the acquired four-involution marking. The saved list of thirty-two indices $(i_1, \ldots, i_{32})$ defines

$$x = r_{i_1} \cdots r_{i_{32}}, \qquad t = r_1.$$

The whole marking, its ambient generators and the named-group identification are bound inputs. For the first five groups the ambient extension is, respectively, `PrimitiveGroup(144,4)`, `PrimitiveGroup(100,2)`, `PrimitiveGroup(100,4)`, `PrimitiveGroup(2058,2)`, and `PrimitiveGroup(1782,2`
Its derived subgroup is the specified simple group. The $J_3 : 2$ representation and defining presentation are acquired from the earlier certificate and its Atlas inputs [3]. We retain those named identifications; we do not charge them as if the group name followed from an order.

The remaining four pairs are literal permutations in `MathieuGroup(11)`, `MathieuGroup(22)`, the derived subgroup of `PrimitiveGroup(275,2)`, and `PrimitiveGroup(266,1)`. The last catalogue record explicitly identifies J_1; that identification is a retained input too. The checker tests membership of both permutations in each named group before comparing the generated subgroup's order with its order.

3 Certificate and scope

For every saved pair, the finite checker verifies $t^2 = 1$, $t \neq 1$, membership in the specified group, generation of the full group, and each of the four orders in the table. Since the group is finite, a contained subgroup of equal order is the whole group. This proves generation from the saved data. Proposition 1 then proves the absence of reversal. The geometric conclusion uses only the retained reconstruction theorem and the already identified simple group.

The witness searches are construction aids and are not replay inputs. No search over candidate pairs or automorphisms occurs in verification. GAP's exact permutation and subgroup algorithms are computational proof infrastructure; they have not been replaced here by Lean proofs. The complete checker, saved data and all required group identifications remain inside the dependency boundary for any later comparison.

The other fourteen sporadic cases and the classical and exceptional Lie families in the source remain separate obligations. This component does not complete the original manuscript. The existing existence results, the two printed Mathieu witnesses, the external group classification and the geometric reconstruction theorem retain their attribution; the new saved witnesses and this common checking argument are the contribution of this component.

References

[1] D. Leemans and M. W. Liebeck, *Chiral polyhedra and finite simple groups*, retained arXiv:1603.07713v3, Sections 1 and 6.

[2] E. Schulte and A. Ivić Weiss, *Chiral polytopes*, Applied Geometry and Discrete Mathematics, DIMACS Series 4 (1991), 493–516.

[3] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker and R. A. Wilson, *Atlas of Finite Groups*, Oxford University Press, 1985.

[4] Human Compression project, *Explicit sporadic markings and their incidence certificates*, accepted N006 component; exact files and admission recorded in the dependency ledger.